

Spring AI Roadmap

Basic to Advanced • Industry-Level Curriculum

Complete roadmap with phases, topics, and subtopics based on the provided curriculum.

PHASE 1 – AI & Generative AI Fundamentals

1.1 Artificial Intelligence

- What is AI?
- Types of AI
- Narrow AI
- Generative AI
- Traditional AI vs Generative AI

1.2 Machine Learning Awareness

- What is ML?
 - Training
 - Inference
 - Model
 - Dataset
 - Features
 - Labels
 - Basic supervised/unsupervised learning
- Only basic awareness is required for Spring AI.

1.3 Deep Learning Awareness

- Neural networks
- Deep learning
- Transformers
- Why transformers became important

1.4 Generative AI

- What is Generative AI?
- Text generation
- Image generation
- Audio generation
- Code generation

1.5 Large Language Models

- What is an LLM?
- LLM architecture — high level
- Pre-training
- Fine-tuning
- Inference

- Model parameters
- Context window

1.6 Tokens

- What is a token?
- Tokenization
- Input tokens
- Output tokens
- Token limits
- Token cost

1.7 Prompts

- Prompt
- System message
- User message
- Assistant response
- Context
- Instructions

1.8 LLM Parameters

- Temperature
- Maximum output tokens
- Top-p
- Model selection

1.9 LLM Limitations

- Hallucination
- Knowledge cutoff
- Context limitations
- Incorrect reasoning
- Prompt injection
- Bias
- Privacy concerns

PHASE 2 – Spring AI Fundamentals

2.1 Introduction to Spring AI

- What is Spring AI?
- Why Spring AI?
- Spring AI architecture
- Spring AI vs direct AI SDK usage
- AI application architecture

2.2 Spring AI Project Setup

- Spring Boot version compatibility
- Spring AI dependencies
- Maven setup
- Gradle setup
- API keys
- Environment variables
- Configuration files

2.3 Spring AI Architecture

Understand:

Spring Boot

Spring AI

Model API

AI Model

Response

2.4 Model Abstraction

- Chat model
- Embedding model
- Image model
- Audio model
- Model providers
- Provider abstraction

2.5 AI Application Configuration

- `application.properties`
- `application.yml`
- API credentials
- Model configuration
- Default options
- Environment-specific configuration

PHASE 3 – ChatClient

3.1 ChatClient Introduction

- What is ChatClient?
- ChatClient architecture
- Creating ChatClient
- Dependency injection

3.2 Basic Chat

- User input
- Sending prompt
- Receiving response
- `.prompt()`
- `.call()`
- `.content()`

3.3 Prompt Construction

- Simple prompt
- Dynamic prompt
- System prompt
- User prompt
- Prompt parameters

3.4 Prompt Templates

- Template variables
- Dynamic values
- Reusable prompts
- Prompt construction

3.5 ChatResponse

- Chat response
- Metadata
- Model information
- Token usage
- Response handling

3.6 Streaming

- Normal response
- Streaming response
- Server-Sent Events
- Streaming REST APIs
- Streaming UI

3.7 Error Handling

- API failures
- Authentication errors
- Rate limits

- Timeout
- Invalid requests
- Exception handling

3.8 ChatClient Applications

- Simple chatbot
- Java tutor
- Interview assistant
- Code explanation assistant

PHASE 4 – AI Model Providers

4.1 OpenAI

- API configuration
- Chat models
- Embeddings
- Model selection

4.2 Google

- Gemini
- Model configuration
- Chat integration

4.3 Anthropic

- Claude
- API configuration
- Model integration

4.4 Azure AI

- Azure model integration
- Configuration

4.5 Amazon Bedrock

- Bedrock basics
- Model access
- Spring AI integration

4.6 Ollama

- Local LLMs
- Installing Ollama
- Running models
- Spring AI + Ollama
- Local development

4.7 Model Switching

Understand:

Application

Spring AI

Provider A

OR

Provider B

OR

Local Model

Learn:

- Provider independence
- Configuration switching
- Model selection

- Fallback strategies

PHASE 5 – Prompt Engineering

5.1 Prompt Fundamentals

- Clear instructions
- Context
- Constraints
- Expected output

5.2 Zero-Shot Prompting

- Definition
- Examples
- Java application

5.3 Few-Shot Prompting

- Examples
- Input/output examples
- When to use

5.4 Role-Based Prompting

- System role
- User role
- Assistant role

5.5 Context Injection

- Dynamic data
- Database information
- User information
- Application context

5.6 Prompt Templates

- Reusable prompts
- Variables
- Dynamic prompts

5.7 Structured Responses

- JSON output
- Schema
- Java records
- POJO mapping

5.8 Prompt Security

- Prompt injection
- Malicious instructions
- Data leakage
- Input validation

5.9 Prompt Optimization

- Reducing unnecessary tokens

- Improving instructions
- Reducing ambiguity
- Output constraints

PHASE 6 – Structured Output

This deserves its own phase because it is extremely useful in real applications.

6.1 Why Structured Output?

- Text vs structured data
- JSON responses
- Application-friendly responses

6.2 Java Object Mapping

- POJO
- Record
- DTO
- JSON → Java object

6.3 Structured Output

- List output
- Object output
- Nested objects
- Enum output

6.4 Practical Applications

- Resume analysis
- Job recommendation data
- Product classification
- Sentiment analysis
- Interview evaluation

PHASE 7 – Embeddings

7.1 Embedding Fundamentals

- What is an embedding?
- Text representation
- Vector representation
- Dimensions

7.2 Semantic Similarity

- Keyword search
- Semantic search
- Similar meaning
- Similarity score

7.3 Embedding Models

- Embedding model
- Input text
- Output vector

7.4 Spring AI EmbeddingModel

- Configuration
- Generating embeddings
- Handling vectors

7.5 Similarity

- Cosine similarity
- Distance
- Similarity ranking

7.6 Practical Applications

- Document search
- Similar questions
- Recommendation systems
- Semantic search

PHASE 8 – Vector Databases

8.1 Vector Database Fundamentals

- What is a vector database?
- Why relational databases alone aren't enough for semantic search
- Vector storage

8.2 Documents

- Document
- Content
- Metadata

8.3 VectorStore

- Spring AI VectorStore
- Adding documents
- Searching documents
- Deleting documents

8.4 PGVector

- PostgreSQL
- PGVector
- Vector column
- Similarity search

8.5 Other Vector Databases

Awareness of:

- MongoDB Atlas
- Redis
- Pinecone
- Qdrant
- Weaviate
- Elasticsearch

8.6 Metadata

- Metadata fields
- Metadata filtering
- User-specific documents
- Department-specific documents

PHASE 9 – RAG

9.1 RAG Fundamentals

- What is RAG?
- Why RAG?
- LLM limitations
- External knowledge

9.2 RAG Architecture

Documents

Chunking

Embeddings

Vector DB

Retrieval

Context

LLM

Answer

9.3 Document Loading

- PDF
- Text
- HTML
- JSON
- Word documents
- Web content

9.4 Document Processing

- Parsing
- Cleaning
- Metadata
- Document objects

9.5 Text Splitting

- Chunking
- Chunk size
- Chunk overlap
- Why chunking matters

9.6 Embedding Pipeline

- Document → chunks
- Chunks → embeddings
- Embeddings → vector database

9.7 Retrieval

- Similarity search
- Top-K results
- Metadata filtering

9.8 Context Injection

- Retrieved context
- Prompt construction
- Context + question

9.9 Spring AI RAG

- QuestionAnswerAdvisor
- RetrievalAugmentationAdvisor
- Retrieval pipeline

9.10 RAG Problems

- Poor chunks
- Irrelevant retrieval
- Missing context
- Too much context
- Hallucination

PHASE 10 – Chat Memory

10.1 Conversation Fundamentals

- Stateless chatbot
- Stateful chatbot
- Conversation history

10.2 Chat Memory

- What is ChatMemory?
- User conversations
- Message history

10.3 Memory Storage

- In-memory
- Persistent memory
- Database-backed memory

10.4 User-Specific Memory

- User ID
- Conversation ID
- Multiple conversations

10.5 Memory + RAG

Understand:

User

Memory

RAG

LLM

Response

10.6 Memory Management

- Context limits
- Summarization
- Old message handling

PHASE 11 – Tool Calling

11.1 Tool Calling Fundamentals

- What is tool calling?
- Function calling
- Why LLMs need tools

11.2 Creating Tools

- @Tool
- Tool methods
- Tool descriptions

11.3 Tool Parameters

- Primitive parameters
- Objects
- Validation
- Input schemas

11.4 Tool Execution

- LLM decides
- Tool invocation
- Tool result
- Final response

11.5 Spring Boot + Tools

Connect tools to:

- Services
- Repositories
- REST APIs
- Databases

11.6 Multiple Tools

Example:

LLM

■■■ Search Product

■■■ Get Order

■■■ Cancel Order

■■■ Check Delivery

11.7 Tool Security

- Authentication
- Authorization
- Input validation
- Sensitive operations
- Human approval

PHASE 12 – Advisors

12.1 Advisor Fundamentals

- What is an Advisor?
- Why Advisors?
- Advisor lifecycle

12.2 Advisor Chain

- Multiple advisors
- Execution order
- Request modification
- Response modification

12.3 Built-in Advisors

- Memory
- RAG
- Tool-related patterns

12.4 Custom Advisors

- Creating custom logic
- Request interception
- Response processing

12.5 Practical Use Cases

- Logging
- Security
- RAG
- Memory
- Prompt modification
- Content filtering

PHASE 13 – MCP

13.1 MCP Fundamentals

- What is MCP?
- Why MCP?
- MCP architecture
- MCP vs traditional APIs

13.2 MCP Components

- MCP Host
- MCP Client
- MCP Server
- Tools
- Resources
- Prompts

13.3 Spring AI MCP

- MCP client
- MCP server
- Configuration
- Tool integration

13.4 MCP Tools

- Exposing tools
- Consuming tools
- Tool discovery

13.5 MCP Resources

- Files
- Database information
- External resources

13.6 MCP Security

- Authentication
- Authorization
- Access control
- Sensitive tools

PHASE 14 – Agentic AI

14.1 Agent Fundamentals

- What is an AI agent?
- Chatbot vs Agent
- Agent architecture

14.2 Agent Loop

User

LLM

Think/Decide

Tool

Result

LLM

Next Action

Final Answer

14.3 Tool-Based Agents

- Tool selection
- Tool execution
- Multiple tools

14.4 Agent Memory

- Short-term memory
- Long-term memory
- Conversation history

14.5 RAG Agents

- Knowledge retrieval
- Retrieval decision
- RAG + tools

14.6 Planning

- Task decomposition
- Multi-step tasks
- Sequential execution

14.7 Human-in-the-Loop

- Approval
- Sensitive operations
- Confirmation

14.8 Agent Security

- Tool permissions
- Prompt injection
- Data access
- Action restrictions

PHASE 15 – Multimodal AI

15.1 Multimodal Fundamentals

- Text
- Image
- Audio
- Video awareness

15.2 Vision

- Image input
- Image understanding
- Image analysis

15.3 PDF/Image AI

- Extracting information
- Document understanding
- Image-based documents

15.4 Speech

- Speech-to-text
- Audio processing

15.5 Text-to-Speech

- Text → voice
- Voice assistant architecture

15.6 Multimodal Applications

- Image analyzer
- Document analyzer
- Voice assistant

PHASE 16 – Production Spring AI

16.1 AI Application Architecture

- Layered architecture
- Service architecture
- AI service layer
- Model abstraction

16.2 Security

- JWT
- User authorization
- API security
- Prompt security
- Tool security

16.3 Rate Limiting

- Request limits
- User limits
- Model limits

16.4 Token Management

- Input tokens
- Output tokens
- Cost tracking
- Token optimization

16.5 Caching

- Response caching
- Embedding caching
- Prompt caching concepts

16.6 Async AI

- Async requests
- `CompletableFuture`
- Background processing

16.7 Streaming

- SSE
- Streaming UI
- Partial responses

16.8 Error Handling

- Timeout
- Rate limit
- Provider failure
- Model failure
- Retry

16.9 Model Fallback

Example:

Primary Model

Failure

Fallback Model

16.10 Observability

- Logging
- Metrics
- Tracing
- AI request monitoring
- Token monitoring

PHASE 17 – Advanced RAG

17.1 Basic RAG

- Retrieve → Generate

17.2 Advanced RAG

- Query transformation
- Query rewriting
- Multi-query retrieval
- Hybrid search

17.3 Retrieval

- Top-K
- Filtering
- Ranking
- Reranking

17.4 Chunking Strategies

- Fixed-size chunks
- Sentence chunks
- Semantic chunks
- Parent-child chunks

17.5 Context Optimization

- Context compression
- Duplicate removal
- Relevant-context selection

17.6 Hybrid Search

Combine:

Keyword Search

Vector Search

Better Retrieval

17.7 RAG Evaluation

- Retrieval accuracy
- Relevance
- Groundedness
- Faithfulness
- Answer quality

PHASE 18 – AI Evaluation

18.1 Why Evaluate AI?

- AI responses are probabilistic
- Output can change
- Quality needs measurement

18.2 Evaluation Types

- Retrieval evaluation
- Response evaluation
- Relevance
- Groundedness

18.3 Test Dataset

- Questions
- Expected context
- Expected answers

18.4 AI Testing

- Prompt testing
- Regression testing
- RAG testing
- Tool testing

18.5 Production Evaluation

- User feedback
- Quality metrics
- Failure analysis

PHASE 19 – Deployment

19.1 Application Packaging

- Maven
- JAR
- Configuration

19.2 Environment Configuration

- Dev
- Test
- Production

19.3 Secrets

- API keys
- Environment variables
- Secret management

19.4 Database

- PostgreSQL
- PGVector
- Production database

19.5 Deployment

- AWS basics
- Cloud deployment
- Managed databases
- Application monitoring

19.6 Production Architecture

Client

Load Balancer

Spring Boot App

Spring AI

LLM PGVector Tools

Database